GSD RESEARCH MISSION PACKAGE — FIVE-PASS DEEP RESEARCH

Forward Error Correction Channel Coding & Reliability Theory

From Shannon's 1948 Theorem to Quantum LDPC Codes and AI-Native Decoders

A Five-Pass Iterative Deep Research Study

Vision → Research → Mission

Full Pipeline Execution — Fleet Activation Profile

Research Passes: Foundations | Code Families | Applications | Distributed/Mesh Systems | Emerging Frontiers

Modules: 6 (Fleet Profile) | **Estimated:** 18–24 context windows across 6–8 sessions

Date: March 27, 2026 | **Status:** Ready for GSD Orchestrator

Author: Miles Tiberius Foxglove / Tibsfox | **Build:** GSD v1.49+ / DACP milestone

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STAGE 1 — VISION DOCUMENT

1 Forward Error Correction — Vision Guide

1.1 Document Metadata

Date: March 27, 2026

Status: Research Complete — Ready for GSD Orchestrator

Depends On: gsd-mesh-prototype-spec, gsd-silicon-layer-spec, fox-infrastructure-group vision docs

Context: Five-pass iterative deep research; six research modules; Fleet activation profile

Ecosystem Role: Foundation layer for GSD Mesh reliability architecture, Fox Infrastructure Group Pacific Spine communications resilience, and GSD-OS local-cloud data integrity guarantees

GSD Build: v1.49+ (DACP milestone branch)

1.2 Vision

There is a geometry to noise. Every signal traveling through the physical world — whether photons threading fiber at the speed of light under the Pacific Ocean, radio waves bouncing through the ionosphere, or bits crawling across a failing NAND flash cell — carries within it an irreducible stochastic error. The channel does not care about your intentions. It corrupts impartially. The fundamental question of information theory is not whether noise exists, but whether meaning can survive it.

Claude Shannon answered this question in 1948 in a paper so profound it retroactively made every communications system before it primitive. His noisy-channel coding theorem established a limit — the channel capacity C — below which reliable communication is *theoretically possible*, and above which it is *provably impossible*. Not "very hard." Not "usually fails." Provably, mathematically impossible. The theorem was existence-only; it proved a reliable channel encoder existed without constructing one. For the next seventy-five years, the entire field of coding theory has been a single sustained engineering effort to approach Shannon's limit with real, implementable codes.

Forward Error Correction (FEC) is the technology family born from this effort. Where the human instinct is to demand a second copy (retransmit), FEC does something more architectural: it weaves redundancy *into* the original transmission so that any sufficiently large fragment of the received signal contains enough geometric information to reconstruct the whole. The receiver does not ask. The receiver *computes*. The redundancy is not overhead to be minimized — it is structural material, like rebar in concrete, giving the message the strength to survive.

The Amiga Principle lurks here at cosmological scale. Nature's most reliable systems — DNA replication with its redundant codon pairs, the human visual cortex with its

massively parallel processing and cross-checking, the immune system with its overlapping pattern recognition — all exploit structured redundancy not as waste but as architectural leverage. FEC is humanity's attempt to engineer this same leverage into silicon and photonics. And the best modern codes — Low-Density Parity-Check (LDPC) codes, Polar codes, Raptor fountain codes — are not brute-force solutions. They are exquisitely minimal: sparse graphs that propagate belief iteratively, polarization transforms that split a noisy channel into two cleaner ones, rateless codes that generate an unlimited stream of encoded symbols so any sufficiently large puddle of received data reconstructs the source. These are Amiga solutions to a Shannon problem.

For the GSD ecosystem and Fox Infrastructure Group, this research mission is not abstract. The Pacific Spine communications backbone — spanning fiber routes from British Columbia to Chile, ocean compute clusters off the Washington coast, mesh radio nodes at tribal sovereign sites, and satellite uplinks for CSZ emergency coordination — will transmit many petabytes of data across channels with widely varying noise profiles, atmospheric turbulence, seismic disruption, and adversarial interference. Every design decision in that infrastructure is implicitly a FEC decision: which codes, what overhead ratios, what decoding latency, what error floor behavior, what behavior at the cliff edge. Getting these decisions right at the architectural level means the difference between a network that degrades gracefully under pressure and one that fails catastrophically at the moment of maximum need.

This mission exists to produce the deepest possible reference package on FEC theory and practice — from Shannon's foundational theorem through the current state of quantum LDPC codes and AI-native neural decoders — so that every GSD infrastructure decision is grounded in rigorous information-theoretic bedrock.

1.3 Problem Statement

1. **The Reliability Gap at Infrastructure Scale.** Modern communication systems routinely operate at aggregate data rates measured in terabits per second across links spanning thousands of kilometers, through media with vastly different noise characteristics: optical fiber (extremely low BER baseline, catastrophic burst failure during earthquakes), free-space optical (atmospheric turbulence causing Poisson-distributed burst losses), 5G mmWave radio (rain fade, blockage fading), and satellite links (significant propagation delay, Doppler shift, ionospheric scintillation). A Pacific Spine network must select, configure, and dynamically adapt FEC schemes across all of these channel types simultaneously, yet the engineering literature is fragmented across standards bodies (ITU-T, IETF, 3GPP, DVB), application domains, and academic research that rarely cross-references. There is no unified reference tailored to the multi-modal, high-stakes infrastructure use case.

2. **The Shannon Gap: Theory vs. Implementation.** Shannon proved in 1948 that codes achieving capacity exist. Gallager invented LDPC codes in 1960 that are now known to approach Shannon capacity. The gap between these theoretical results and deployed systems has been closing for 75 years, but the "last mile" of the Shannon gap — the difference between the best known practical codes and the theoretical limit — remains consequential in energy-constrained deployments, bandwidth-limited deep-space links, and latency-sensitive real-time systems. Un-

derstanding precisely how each code family performs relative to the Shannon limit, and what tradeoffs are incurred in approaching it, is essential for infrastructure design.

3. **The Quantum Transition Problem.** Quantum computing, if it matures as trajectory suggests, will break RSA and elliptic-curve cryptography — but it will also create a massive new demand for quantum error correction (QEC), which is the quantum analog of FEC. LDPC codes are already being adapted into quantum LDPC (qLDPC) codes for IBM's quantum processors. Google's Willow chip demonstrated practical QEC in December 2024. Meanwhile, classical FEC research is being cross-fertilized with quantum coding theory in ways that are advancing both fields. The Fox Infrastructure Group's FoxCompute ocean data centers will eventually need to service quantum workloads; understanding the FEC/QEC convergence now is architecturally critical.

4. **The AI-Native Decoder Inflection.** Deep neural networks are now being applied to FEC decoding problems, in some cases approaching or exceeding maximum-likelihood decoder performance at reduced computational cost and with explicit hardware acceleration pathways. NASA's JPL has active research programs applying DNNs to quantum LDPC decoding. The 6G standards process (targeting 2030 deployment) is evaluating AI-native channel coding. For a GSD ecosystem built around AI-assisted execution, the convergence of FEC and neural networks is not a curiosity but a core architectural opportunity.

5. **The Mesh Reliability Architecture Gap.** Traditional FEC is point-to-point: one encoder, one decoder, one channel model. The GSD Mesh prototype and Fox Infrastructure Group Pacific Spine require multi-hop, multi-path, multicast-capable FEC that maintains reliability semantics across complex network topologies. Distributed FEC (network coding, fountain codes in multi-hop topologies, cross-layer FEC adaptation in SD-WAN fabrics) is a distinct and underexplored engineering domain that existing single-link FEC literature does not address. The DACP agent communication protocol itself needs FEC-informed reliability semantics for its structured three-part bundles.

1.4 Core Concept

The Shannon Machine: FEC transforms unreliable channels into reliable ones by exploiting the geometry of code spaces — adding structured redundancy such that the minimum distance between valid codewords exceeds the maximum expected noise magnitude, enabling the decoder to map any received word to its nearest valid codeword with high probability.

Formally, for a discrete memoryless channel (DMC) with capacity C bits per channel use, Shannon's second theorem states: for any rate $R < C$ and any $\epsilon > 0$, there exists a block code of sufficient length n such that the block error probability $P_e < \epsilon$. Conversely, for $R > C$, reliable communication is impossible regardless of code complexity. The code rate is defined as $R = k/n$ where k is the number of information bits and n is the total codeword length. The redundancy overhead is $(n - k)/k$, and the minimum Hamming distance d_{min} of a code determines its error-correcting power: a code can correct up to $t = \lfloor (d_{\text{min}} - 1)/2 \rfloor$ errors per codeword.

1.4.1 Domain Architecture Map

```
FEC RESEARCH DOMAIN ARCHITECTURE ====================================
INFORMATION THEORY FOUNDATIONS +---------------------------------+ |
Shannon (1948): C = B*log2(1+S/N)| | Hamming (1950): distance metric |
| Gallager (1960): sparse graphs | +---------------------------------+
| +---------+---------+ | | CLASSICAL FEC QUANTUM FEC (Block
Conv.) (Stabilizer) | | +--+--+--+--+ +--+--+--+--+ |Ha|RS|BC|Tu|
|Su|To|qL|Ne| |mm| |H |rb| |rf|ri|DP|ur| |in| | |o | |ac|c |C
|al| +--+--+--+--+ +--+--+--+--+ | | | LDPC/Polar Fountain | | |
| +-----+---------+---------+ | DECODING ALGORITHMS | | Viterbi /
Belief Prop. | | Sum-Product / SC / MAP | | Neural / ML Decoders |
+-----+---+----------------+ | | +-----+---+-----------+ | APPLICATION
LAYER | | 5G NR | Optical | | Space | Storage | | IoT | SD-WAN | | Mesh
| Streaming | +-------+-------------+ | +-----+---------------------+ |
FOX INFRASTRUCTURE GROUP | | Pacific Spine FEC Layer | | DACP Reliability
Overlay | | CSZ Emergency Resilience | +---------------------------+
```

1.4.2 Module Architecture

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6-MODULE FLEET EXECUTION PLAN ==============================
Module 01: THEORETICAL FOUNDATIONS - Shannon theorem, channel models,
Hamming distance - Code rate / capacity tradeoff mathematics - Coding
gain, BER curves, SNR relationships [CRAFT-THEORY, Opus] [sequential, Wave
1A]
Module 02: CODE FAMILIES TAXONOMY - Block codes: Hamming, BCH,
Reed-Solomon - Convolutional codes: rate, constraint length, trellis -
Modern codes: Turbo, LDPC, Polar, Fountain/Raptor [CRAFT-CODE, Sonnet]
[parallel with 03, Wave 1A]
Module 03: DECODING ALGORITHMS - Hard vs. soft decision decoding -
Viterbi algorithm, MAP/BCJR - Belief propagation, sum-product, min-sum
- Successive cancellation, SC-list decoding - Neural network decoders
(emerging) [CRAFT-DECODE, Sonnet] [parallel with 02, Wave 1A]
Module 04: APPLICATION DOMAINS - Wireless: 5G NR (LDPC + Polar), WiFi, IoT
- Optical: submarine cables, coherent DWDM - Space: Voyager Reed-Solomon,
deep space turbo - Storage: NAND Flash, QLC FEC requirements - Streaming:
Raptor/RaptorQ, IPTV, ATSC 3.0 - SD-WAN / mesh: distributed FEC topology
[CRAFT-APP, Sonnet] [Wave 1B, after Mod 02+03]
Module 05: EMERGING FRONTIERS - Quantum error correction: surface codes,
QLDPC - AI/ML decoders: neural FEC, DNN channel coding - 6G channel coding
research directions - Information-theoretic security (wiretap channels)
[CRAFT-FRONTIER, Opus] [Wave 1B, parallel with 04]
Module 06: GSD/FOX INFRASTRUCTURE INTEGRATION - FEC selection guide for
Pacific Spine segments - DACP reliability layer design with FEC semantics
- GSD Mesh prototype FEC architecture - CSZ emergency mesh: fountain codes
for disrupted topology [CRAFT-GSD, Opus] [Wave 2 synthesis, depends on
01-05]
```

1.5 Research Modules

1.5.1 Module 01: Theoretical Foundations

This module establishes the bedrock mathematics of FEC. It begins with Shannon's 1948 noisy-channel coding theorem and the channel capacity formula $C = B\log_2(1+$

S/N), proceeds through Hamming's 1950 formalization of error-correcting codes and the Hamming distance metric, and builds up the complete mathematical framework of code rate, minimum distance, coding gain, and the Shannon limit. The module also covers channel models: the binary symmetric channel (BSC), binary erasure channel (BEC), additive white Gaussian noise (AWGN) channel, and the burst-error channel. It defines net coding gain (NCG), bit error rate (BER) curves as a function of E_b/N_0, and establishes the cliff effect — the all-or-nothing failure mode when the channel quality falls below the code's design operating point.

1.5.2 Module 02: Code Families Taxonomy

A comprehensive survey of every major FEC code family, organized chronologically and architecturally. Classic block codes: Hamming (7,4) through extended Hamming; BCH codes (Bose-Chaudhuri-Hocquenghem); Reed-Solomon codes as non-binary BCH codes operating on symbol alphabets. Classic convolutional codes: rate k/n with constraint length K, trellis diagrams, punctured convolutional codes. The revolutionary codes of the 1990s–2000s: Turbo codes (Berrou et al., 1993) using recursive systematic convolution with interleaving; LDPC codes (Gallager 1960, rediscovered 1996) as sparse graph codes with belief-propagation decoding; Polar codes (Arikan, 2008) exploiting channel polarization to synthesize noiseless and pure-noise synthetic channels. Fountain/Rateless codes: LT codes (Luby, 2002), Raptor codes (Shokrollahi, 2001/2004), and RaptorQ (IETF RFC 6330). Concatenated and product codes: serially concatenated codes, turbo product codes (TPC), staircase codes for optical transport.

1.5.3 Module 03: Decoding Algorithms

The decoder is at least as important as the code itself. This module surveys the complete landscape of decoding strategies. Hard-decision decoding: syndrome computation, lookup tables, bounded-distance decoding. Soft-decision decoding and the ~2–3 dB gain it provides. The Viterbi algorithm as maximum-likelihood sequence estimation on convolutional codes. The BCJR/MAP algorithm for turbo decoding. Belief propagation (sum-product algorithm) on Tanner graphs for LDPC decoding. The min-sum approximation and its hardware efficiency. Successive cancellation (SC) decoding for polar codes and the SC-list improvement. The error floor phenomenon in LDPC codes and trapping set analysis. Hardware implementations: pipeline parallelism, quasi-cyclic LDPC for VLSI efficiency. The emerging frontier of neural network decoders: hypernetworks, recurrent decoders, and learned belief propagation.

1.5.4 Module 04: Application Domains

FEC does not exist in a vacuum — each application domain imposes unique constraints on code selection, code rate, overhead ratio, latency budget, hardware complexity, and decoding power consumption. This module traces FEC deployment across eight major application sectors. Wireless communications evolution: convolutional codes in 2G GSM, Turbo codes in 3G UMTS and 4G LTE, LDPC codes for data channels and Polar codes for control channels in 5G NR (3GPP TS 38.212). Optical fiber transport: ITU-T G.709 and G.975 Reed-Solomon GFEC at 7% overhead, hard-decision staircase FEC at 20% overhead, soft-decision LDPC at 25% overhead for coherent 100G/400G/800G submarine and metro networks. Deep space: NASA JPL's concatenated RS + convolutional codes on Voyager, turbo codes on Mars Reconnaissance

Orbiter, and the emerging CCSDS standards for the Lunar Gateway and Artemis communications backbone. Flash memory and SSDs: BCH then LDPC migration as NAND process nodes shrink and raw BER climbs. Streaming and broadcast: Raptor/RaptorQ in 3GPP MBMS and ATSC 3.0 Next Gen TV. SD-WAN and enterprise mesh: packet-layer FEC for VoIP, video conferencing, and real-time control over impaired WAN links.

1.5.5 Module 05: Emerging Frontiers

The frontier of FEC is being reshaped by three simultaneous forces: the emergence of viable quantum hardware requiring quantum error correction, the application of machine learning to decoder design, and the 6G standardization process that must decide what comes after LDPC+Polar. Quantum error correction (QEC): the stabilizer code formalism, surface codes (topological codes), quantum LDPC codes (qLDPC), and the 2024–2025 demonstration of practical QEC on IBM and Google hardware. Google's Willow chip and IBM's qLDPC transition. The 120+ peer-reviewed QEC papers published in 2025. AI-native FEC: neural network decoders for classical codes, reinforcement learning for adaptive code rate selection, and DNN-assisted channel estimation interleaved with FEC decoding. 6G research: polar-RS concatenation for ultra-reliability, semantic coding as a paradigm shift beyond the symbol level, and the 2025 Nature paper on concatenated RS + polar codes for 6G candidate evaluation.

1.5.6 Module 06: GSD / Fox Infrastructure Integration

The synthesis module that translates all five research tracks into actionable architecture for the GSD ecosystem and Fox Infrastructure Group. This module produces: (1) a FEC selection guide mapping Pacific Spine channel segments to recommended code families and overhead ratios; (2) a DACP reliability layer specification that incorporates FEC semantics into the three-part agent communication bundle (intent + data + executable code), enabling agent-to-agent messages to survive partial context window corruption; (3) a GSD Mesh prototype FEC architecture specification for the multi-hop mesh topology across Cascadia nodes; (4) a Cascadia Subduction Zone (CSZ) emergency mesh specification using fountain/Raptor codes to maintain communication reliability under partial topology disruption when conventional routing fails.

1.6 Chipset Configuration

```
CHIPSET YAML --- FEC DEEP RESEARCH MISSION ==========================================
chipset: agnus:  Orchestration / DMA / context model: claude-opus-4-6
role: > Mission orchestration across 6 modules. Manages wave transitions,
cache policy, cross-module consistency, GSD integration synthesis (Module
06). Approves all infrastructure recommendations before publication wave.
token_budget : 80000 waves : [0, 2, 3]
denise:  Display / output rendering model: claude-sonnet-4-6 role:
> Document assembly, LaTeX compilation, table generation, figure
composition, index.html production, bibliography formatting. Runs Wave 3
publication. token_budget : 40000 waves : [3]
paula:  I/O / anticipatory processing model: claude-haiku-4-5 role: >
Shared schema generation, template scaffolding, bibliography validation,
cross-reference indexing. Wave 0 only. token_budget : 8000 waves : [0]
craft_theory : Module01 specialistmodel : claude - opus - 4 - 6 role :
Shannontheorem, channelcapacitymath, Hammingdistance, codinggainderivations.Requiresdeepmathematicalprecision
30000 wave : 1A
craft_code : Module02 specialistmodel : claude - sonnet - 4 - 6 role :
Codefamiliessurvey, parametertables, performancecharacterizations, historicaltimelineconstruction.token_budget :
30000 wave : 1A
craft_decode : Module03 specialistmodel : claude - sonnet - 4 - 6 role :
Decodingalgorithmsurvey, hardwareimplementationpatterns, errorflooranalysis, neuraldecoderresearch.token_budget :
30000 wave : 1A
craft_app : Module04 specialistmodel : claude - sonnet - 4 -
6 role : Applicationdomainsurveyacross8sectors, standardscross -
reference, deploymentcasestudies.token_budget : 35000 wave : 1B
craft_frontier : Module05 specialistmodel : claude - opus - 4 - 6 role :
Quantumerrorcorrection, AIdecoders, 6Gresearch.Requiresfrontierresearchsynthesiswithappropriateuncertainty.toke
35000 wave : 1B
craft_gsd : Module06 synthesismodel : claude - opus - 4 - 6 role :
GSD/FoxInfrastructureintegration.TranslatesFECresearchintoactionablearchitectureforPacificSpine, DACP, GS1
40000 wave : 2
gary_68000 : CPU/gluelogicmodel : claude - haiku - 4 - 5 role :
Configurationmanagement, toolcalls, fileoperations, compilationautomation.token_budget :
5000 waves : [0, 1A, 1B, 2, 3]
```

1.7 Scope Boundaries

1.7.1 In Scope (v1.0)

- Complete mathematical foundations of FEC from Shannon (1948) through current state of the art
- All major classical code families: Hamming, BCH, Reed-Solomon, convolutional, turbo, LDPC, polar, fountain/Raptor
- Core decoding algorithms: Viterbi, BCJR/MAP, belief propagation, sum-product, successive cancellation, SC-list
- Eight major application domains with standards cross-references
- Quantum error correction survey: stabilizer codes, surface codes, qLDPC, 2024–2025 hardware demonstrations

- AI/neural decoder research and 6G coding research directions
- FEC selection architecture for Fox Infrastructure Group Pacific Spine
- DACP reliability overlay specification with FEC semantics
- GSD Mesh prototype FEC architecture
- CSZ emergency mesh fountain code specification
- Complete bibliography with peer-reviewed and standards-body sources

1.7.2 Out of Scope

- Cryptographic error correction and post-quantum cryptography (separate mission)
- Full VLSI implementation blueprints for custom FEC silicon (requires TSMC/Intel Foundry engagement)
- Proprietary code families under active patent protection without published technical specifications
- ARQ (automatic repeat request) and HARQ in detail beyond their relationship to FEC and hybrid schemes
- Non-binary LDPC codes beyond survey level
- Specific procurement decisions for Fox Infrastructure Group hardware (pending vendor engagement)

1.8 Success Criteria

SC-1: Shannon channel capacity theorem stated mathematically with all variables defined, and the tradeoff between code rate and BER explicitly characterized with quantitative examples.

SC-2: All seven major code families (Hamming, BCH, RS, convolutional, turbo, LDPC, polar) characterized with: code parameters (n, k, d_{min}), error-correcting capability t, code rate, typical overhead, and primary application domain.

SC-3: Fountain/Raptor codes documented with: rateless property, LT vs. Raptor vs. RaptorQ performance, IETF RFC 6330 specification reference, and 3GPP/ATSC 3.0 deployment context.

SC-4: Decoding algorithms table mapping each code family to its canonical decoder, decoder complexity class, and hard vs. soft decision behavior.

SC-5: 5G NR channel coding standard documented: LDPC for data channels with base graph selection criteria, Polar for control channels with CA-polar construction, and KP4 FEC for PAM4 SerDes interfaces.

SC-6: Deep space FEC timeline documented: Voyager/Pioneer concatenated codes, Galileo emergency repair, Mars missions turbo codes, and CCSDS current recommendations.

SC-7: Optical transport FEC generations documented: G-FEC (RS 255,239), E-FEC (RS enhanced), EFEC (LDPC), and soft-decision FEC with NCG values in dB.

SC-8: Quantum error correction documented: stabilizer formalism, surface code threshold, Google Willow result (December 2024), IBM qLDPC transition, and at least 3 QEC code families characterized.

SC-9: AI/neural decoder research documented with at least 3 research results, including JPL NASA DNN decoder work and performance comparison to classical decoders.

SC-10: Fox Infrastructure Group FEC selection guide produced with minimum 6 Pacific Spine segment types mapped to recommended code families, overhead ratios, and decoder latency budgets.

SC-11: DACP reliability layer specification produced: at minimum a conceptual architecture showing how FEC semantics protect the three-part bundle (intent + data + executable code) across agent-to-agent hops.

SC-12: CSZ emergency mesh fountain code specification produced: minimum topology failure scenarios, Raptor/RaptorQ configuration recommendation, and expected recovery time under partial mesh disruption.

1.9 Relationship to Other Documents

Document	Relationship	Integration Point
GSD Mesh Prototype Spec	Downstream consumer	Module 06 FEC architecture feeds directly into mesh protocol spec
GSD Silicon Layer Spec	Parallel / complementary	Silicon layer will implement FEC codecs; shares decoder hardware spec
Fox Infrastructure Group Plan	Parent mission	Pacific Spine segment FEC selection (Module 06, SC-10)
DACP Protocol Spec	Downstream consumer	DACP reliability overlay uses fountain code semantics for agent bundles
GSD-OS Local Cloud Provisioning	Downstream consumer	Storage FEC selection for local-cloud data integrity
CSZ Emergency Preparedness	Downstream consumer	Emergency mesh fountain code specification (SC-12)
The Space Between (textbook)	Philosophical parent	L-systems, information theory layer (Ch. 7–8), Complex Plane of Experience

1.10 The Through-Line

The central insight of FEC is not mathematical — it is ontological. In *The Space Between*, the argument runs that meaning lives not in the nodes of a system but in the connections between them. Shannon's theorem says something similar, in the language of bits: the capacity of a channel is not a property of the transmitter, nor of the receiver, but of the relationship between them, mediated by noise. The code is not the data. The code is the structure that gives the data survivability in a world that does not care whether it arrives.

Every modern code family is a different answer to the question: what geometric struc-

ture best exploits the space between valid messages? Turbo codes interleave two recursively-redundant representations of the same data and let them converge iteratively. LDPC codes create a sparse bipartite graph of constraints and propagate beliefs along the edges until the noisy word resolves into a valid codeword. Polar codes split the channel into its extremes — the worst possible subchannel and the best — and place data only on the good side. Raptor codes produce a digital fountain: an infinite stream of drops, any sufficiently large collection of which reconstructs the source. In each case, the architecture is the reliability. There is no brute force. There is only structure.

This is the Amiga Principle at the most fundamental layer of communication: not more power, but better design. Not louder signal, but smarter redundancy. For the GSD ecosystem and Fox Infrastructure Group, this principle is not a design aspiration. It is an engineering specification.

STAGE 2 — RESEARCH REFERENCE

2 Forward Error Correction — Research Reference

2.1 How to Use This Document

This research reference contains detailed findings, quantitative data, and sourced citations for all six research modules. Agents executing research missions should treat this section as the primary factual authority; all claims in Stage 3 deliverables must be traceable to entries here or to the source bibliography.

Key source organizations:

- **IEEE Xplore / IEEE Transactions on Information Theory** — primary peer-reviewed venue for FEC research
- **IETF (Internet Engineering Task Force)** — application-layer FEC standards (RFC 5053, RFC 6330 RaptorQ)
- **ITU-T (International Telecommunication Union)** — optical transport FEC standards (G.709, G.975, G.709.2)
- **3GPP (3rd Generation Partnership Project)** — wireless FEC standards (TS 38.212 for 5G NR)
- **DVB Project** — broadcast FEC standards (DVB-S2 LDPC, DVB-H Raptor)
- **NASA/CCSDS (Consultative Committee for Space Data Systems)** — deep space FEC recommendations
- **Bell System Technical Journal** — Shannon (1948) and Hamming (1950) original papers
- **MIT / Caltech** — Gallager (1960) LDPC dissertation; Arikan (2008) polar codes
- **National Institute of Information and Communications Technology (NICT)** — ground-to-satellite FEC demonstrations (2025)
- **Riverlane / IBM / Google** — quantum error correction hardware results (2024–2025)

2.2 Module 01 Reference: Theoretical Foundations

2.2.1 Shannon's Foundational Theorem (1948)

Claude Shannon published "A Mathematical Theory of Communication" in the *Bell System Technical Journal* in 1948. This single paper established the entire theoretical framework for digital communications and information theory. The core results relevant to FEC are:

The Noisy Channel Coding Theorem: For a discrete memoryless channel (DMC) with capacity C, and for any rate $R < C$ and error probability target $\epsilon > 0$, there exists a block code of length n (sufficiently large) such that the maximum probability

of decoding error $P_e < \epsilon$. For rates $R > C$, no such code exists. The capacity of the AWGN channel (Additive White Gaussian Noise) is:

$$C = B \log_2\left(1 + \frac{S}{N}\right) \text{ bits/second}$$

where B is the bandwidth in Hz, S is signal power, and N is noise power.

Key practical implication: Shannon's theorem is an existence proof. It guarantees that reliable codes exist but does not construct them. The proof relies on random Gaussian codes, which are not computationally implementable. The entire subsequent history of coding theory is the engineering effort to find constructive, efficiently decodable codes that achieve Shannon capacity.

The Shannon limit for AWGN at capacity: For a binary-input AWGN channel at $P_e \to 0$, the minimum required E_b/N_0 (energy per information bit to noise spectral density ratio) approaches -1.59 dB (the Shannon limit). In practice, LDPC codes and turbo codes achieve within 0.1–0.5 dB of this limit with modern decoder implementations.

2.2.2 The Hamming Distance and Code Design

Richard Hamming published "Error Detecting and Error Correcting Codes" in the *Bell System Technical Journal* in 1950, inventing the first practical error-correcting code. The foundational concepts:

Hamming Distance: $d_H(x, y)$ = number of positions in which bit vectors x and y differ.

Minimum Distance of a Code: $d_{\text{min}} = \min_{c_1 \neq c_2} d_H(c_1, c_2)$ over all pairs of distinct codewords c_1, c_2.

Error Detection and Correction Capacity:

- A code with minimum distance d_{min} can **detect** up to $d_{\text{min}} - 1$ errors per codeword
- A code with minimum distance d_{min} can **correct** up to $t = \lfloor (d_{\text{min}} - 1)/2 \rfloor$ errors per codeword
- A code that maximizes d_{min} for given parameters (n, k) is **optimal**; codes achieving the Hamming bound exactly are called **perfect codes**

The Hamming (7,4) Code: The first practical FEC code. Encodes 4 information bits into 7-bit codewords using 3 parity bits. Parameters: $n = 7$, $k = 4$, $d_{\text{min}} = 3$, $t = 1$. Can correct any single-bit error. Code rate $R = 4/7 \approx 0.571$. The generator matrix G and parity check matrix H provide systematic encoding; syndrome computation at the decoder identifies and corrects the single error.

2.2.3 Code Rate and Overhead Tradeoffs

Code Rate	Overhead	NCG Range	Complexity	Typical Use
$R = 1$ (none)	0%	0 dB	None	Perfect channel
$R = 0.93$ (7%)	7%	5–6 dB	Low	Optical G-FEC
$R = 0.87$ (15%)	15%	7–8 dB	Medium	Enhanced optical
$R = 0.80$ (25%)	25%	9–11 dB	High	SD-FEC optical
$R = 3/4$	33%	8–9 dB	Medium	5G NR LDPC
$R = 1/2$	100%	10–11 dB	High	Turbo / deep space
$R = 1/3$	200%	11–12 dB	Very high	Deep space extreme

2.2.4 Net Coding Gain (NCG)

NCG is defined as the improvement in received optical sensitivity (or effective SNR) provided by the FEC code, measured in dB at a target post-FEC BER (typically 10^{-13} to 10^{-15} for optical transport). NCG is calculated as:

$$\text{NCG} = E_b/N_0\,(\text{uncoded}) - E_b/N_0\,(\text{coded})$$

at the target BER, after accounting for the bandwidth expansion due to the overhead. Shannon's NCG limit for a given overhead represents the theoretical maximum NCG achievable by any code. Modern soft-decision LDPC codes approach within 0.5–1.0 dB of this limit.

2.2.5 The Cliff Effect

FEC systems exhibit a threshold behavior called the "cliff effect": above the design pre-FEC BER threshold, the post-FEC BER improves exponentially (waterfall region). Below the threshold (worse channel), the decoder fails and the post-FEC BER approaches the uncoded BER (cliff edge). This threshold behavior becomes more pronounced as codes approach the Shannon limit (stronger codes have steeper waterfall slopes but also more abrupt cliff edges). The cliff effect has critical implications for infrastructure design: systems must be designed with enough link margin to remain above the pre-FEC BER threshold under worst-case channel conditions.

2.3 Module 02 Reference: Code Families Taxonomy

2.3.1 Historical Timeline

Year	Code / Event	Inventor(s)	Significance
1948	Channel capacity theorem	Claude Shannon	Theoretical foundation; proves reliable codes exist
1950	Hamming (7,4) code	Richard Hamming	First practical single-error-correcting code
1955	Binary Erasure Channel	Peter Elias (MIT)	Erasure channel model; Raptor code precursor
1959	BCH codes	Hocquenghem; Bose & Ray-Chaudhuri	Multi-error-correcting binary block codes
1960	Reed-Solomon codes	Irving Reed, Gus Solomon	Non-binary, burst-error-correcting; MDS property
1960	LDPC codes	Robert Gallager (MIT)	Near-Shannon limit; ignored for 36 years
1967	Viterbi algorithm	Andrew Viterbi	Efficient ML decoding of convolutional codes
1974	BCJR / MAP algorithm	Bahl, Cocke, Jelinek, Raviv	Foundation of turbo decoder; soft-output decoding
1993	Turbo codes	Berrou, Glavieux, Thitimajshima	Near-Shannon limit; revitalized coding research
1996	LDPC rediscovery	MacKay, Neal	Belief propagation decoding; modern LDPC era begins
2002	LT codes	Michael Luby	First practical fountain codes
2001/20(	Raptor codes	Amin Shokrollahi	Fountain codes with linear-time encoding/decoding
2008	Polar codes	Erdal Arikan	First codes proven to achieve Shannon capacity; 5G NR adoption
2011	RaptorQ (RFC 6330)	IETF	Enhanced Raptor; ATSC 3.0 standard FEC
2024/25	Quantum LDPC (qLDPC)	IBM / Google / Riverlane	Practical QEC on real quantum hardware

2.3.2 Block Code Family Parameters

Code Family	(n, k) Example	$d_{\min}$	t (correct)	Rate	Key Property
Hamming (7,4)	$(7, 4)$	3	1	0.571	Single-error-correcting; SEC-DED with parity
BCH (31,16)	$(31, 16)$	7	3	0.516	Multi-error-correcting; binary GF construction
Reed-Solomon RS(255,239)	$(255, 239)$	17	8	0.937	Non-binary; MDS; burst-error immunity
Reed-Solomon RS(255,223)	$(255, 223)$	33	16	0.875	CD audio / NASA standard; 16-symbol correction
Reed-Solomon KP4 RS(544,514)	$(544, 514)$	31	15	0.945	PAM4 SerDes; 200/400GbE standard
LDPC (1944, 1296)	$(1944, 1296)$	$\gg$	many	0.667	802.11n WiFi LDPC; iterative BP decoding
LDPC DVB-S2	$(64800, k)$	high	many	var	Satellite TV; 7–8 dB NCG; 0.5–9/10 rates
Polar (1024, 512)	$(1024, 512)$	–	–	0.5	5G NR control channel; provably capacity-achieving

2.3.3 Reed-Solomon Codes: The Workhorse of Reliability

Reed-Solomon (RS) codes are non-binary BCH codes that operate on m-bit symbols over a Galois field GF(2^m). An RS(n, k) code with symbol size m bits has:

- Block length $n = 2^m - 1$ symbols (in GF(2^m)); $n = 255$ for $m = 8$ (most common)
- Number of parity symbols $r = n - k$; capable of correcting $t = r/2$ symbol errors
- Minimum distance $d_{\min} = r + 1 = n - k + 1$ (maximum possible for a linear code — RS is MDS)
- Exceptionally effective against burst errors: a b-bit burst corrupts at most $\lceil b/m \rceil$ symbols

RS codes have been the dominant industrial FEC workhorse for 40+ years: NASA's Voyager mission (RS(255,223)), CD/DVD/Blu-ray disc error correction, QR code recovery, optical transport (G-FEC RS(255,239)), and Ethernet KP4 FEC (RS(544,514) for PAM4 400GbE).

2.3.4 LDPC Codes: The Near-Shannon Standard

Low-Density Parity-Check (LDPC) codes, invented by Robert Gallager at MIT in 1960 and rediscovered by MacKay and Neal in 1996, are defined by a sparse parity-check matrix H where the density of 1s is very low. This sparsity enables the sum-product (belief propagation) decoding algorithm to operate efficiently on the bipartite Tanner graph representation.

Key properties:

- Can be designed to approach the Shannon capacity on AWGN channels to within 0.0045 dB (capacity-approaching sequences shown by Richardson, Shokrollahi, and Urbanke in 2001)
- No inherent minimum distance limitations (unlike turbo codes); better performance at high code rates ($R \geq 3/4$) and lower error floors than turbo codes
- Quasi-cyclic LDPC (QC-LDPC) structures enable highly parallel hardware implementation with low routing complexity
- Used in: 5G NR data channels (3GPP TS 38.212 base graphs BG1 and BG2), 10GBase-T Ethernet, Wi-Fi 802.11n/ac/ax, DVB-S2 satellite, DOCSIS 3.1, and optical transport SD-FEC

2.3.5 Polar Codes: Proven Capacity Achievement

Erdal Arikan published the polar code construction in 2008, proving for the first time that a constructive coding scheme with efficient (polynomial-time) encoding and decoding can achieve the symmetric capacity of any binary-input DMC. The key mechanism is *channel polarization*: applying n copies of a channel through a specific recursive butterfly transform splits the n channels into two groups as $n \to \infty$ — a fraction approaching C of the channels become noiseless ("good channels"), and a fraction approaching $1 - C$ become completely noisy ("bad channels"). Information bits are placed on good channels; frozen bits (known to the receiver) on bad channels.

Practical polar codes use:

- **Successive Cancellation (SC) decoding:** $O(n \log n)$ complexity; relatively low performance at short block lengths
- **SC-List (SCL) decoding with CRC (CA-Polar):** Significant performance improvement; used in 5G NR
- 5G NR uses polar codes for control channels (PBCH, PDCCH, PUCCH) with CRC-aided list decoding; LDPC for data channels (PDSCH, PUSCH)

2.3.6 Turbo Codes: The 1993 Revolution

Berrou, Glavieux, and Thitimajshima presented turbo codes at ICC 1993, demonstrating for the first time near-Shannon-limit error correction in a practical code — achieving 10^{-5} BER at only 0.5 dB from the Shannon limit for a rate-1/2 code at that time. Turbo codes use:

- Two parallel recursive systematic convolutional (RSC) encoders separated by an interleaver

- Iterative decoding using two soft-input/soft-output (SISO) MAP decoders exchanging extrinsic information ("turbo effect")
- BCJR/MAP or Log-MAP algorithm for each constituent decoder
- Performance improves with more decoding iterations (typically 6–8 for good latency/performance tradeoff)
- Primary weakness: error floor at high E_b/N_0 due to low-weight codewords; worse than LDPC at higher code rates
- Used in 3G UMTS, 4G LTE, deep space (CCSDS), and satellite communications

2.3.7 Fountain and Raptor Codes: Rateless Reliability

Fountain codes (rateless erasure codes) represent a fundamentally different design philosophy: rather than transmitting a fixed set of n encoded symbols for k source symbols, the encoder can produce an *unlimited* stream of encoded symbols on demand. The receiver collects any $k' \approx k(1+\epsilon)$ symbols (from any subset of transmitted symbols) and can recover the original k source symbols with high probability.

LT codes (Luby Transform, Michael Luby, 2002): The first practical fountain codes. Each encoded symbol is the XOR of a pseudo-randomly chosen set of source symbols, with the degree (number of source symbols XORed) chosen from the Robust Soliton Distribution. Decoded by belief propagation (iterative message passing). Linear-time encoding and decoding achievable with properly tuned degree distributions.

Raptor codes (Amin Shokrollahi, 2001/2004): Two-stage architecture: (1) a fixed-rate "pre-code" (typically LDPC or HDPC — high-density parity-check) generates intermediate symbols from source symbols, providing a regularization that prevents decoding failure; (2) LT encoding of the intermediate symbols. Raptor codes achieve linear-time encoding/decoding with reception overhead approaching zero.

RaptorQ (IETF RFC 6330, 2011): The most advanced standardized fountain code. Parameters:

- Supports up to 56,403 source symbols per source block
- Up to 16,777,216 encoded symbols generable per source block
- Recovery of source block from any k received symbols with very high probability; typical reception overhead $\epsilon < 0.01$ (i.e., need at most 1% more symbols than k)
- Average overhead converges to approximately $2/k$ additional symbols (for large k, essentially zero overhead)
- Specified in ATSC 3.0 (Next Gen TV) A/331 for broadcast video streaming and file delivery
- Used in 3GPP eMBMS (enhanced Multimedia Broadcast Multicast Service) for mobile TV

2.4 Module 03 Reference: Decoding Algorithms

2.4.1 Decoder Taxonomy

Algorithm	Code Family	Complexity	Soft/Hard	Key Property
Lookup table	Hamming, short codes	$O(1)$	Hard	Syndrome lookup; negligible latency
Berlekamp-Massey	Reed-Solomon	$O(n^2)$	Hard	Berlekamp-Massey for error locator polynomial
Viterbi	Convolutional	$O(n \cdot 2^K)$	Soft	MLE for convolutional; exponential in constraint length K
BCJR / MAP	Turbo constituent	$O(n \cdot \lvert\text{states}\rvert)$	Soft	Optimal soft-output; foundation of turbo decoder
Log-MAP	Turbo	$O(n \cdot \lvert\text{states}\rvert)$	Soft	Numerically stable Log-MAP; standard turbo impl.
Max-Log-MAP	Turbo	$O(n \cdot \lvert\text{states}\rvert)$	Soft	Approximation of Log-MAP; hardware-friendly
Sum-Product (BP)	LDPC	$O(n \cdot d_c \cdot d_v)$	Soft	Iterative message passing on Tanner graph
Min-Sum	LDPC	$O(n \cdot d_c)$	Soft	Approximation of sum-product; hardware-efficient
Successive Cancel.	Polar	$O(n \log n)$	Soft	Sequential; low complexity but poor short-block perf.
SC-List (SCL)	Polar (CA-Polar)	$O(Ln \log n)$	Soft	L candidate paths; 5G NR uses $L = 8$
Belief Prop.	LDPC / general	$O(\text{edges} \cdot I)$	Soft	Converges in I iterations; general-purpose
Neural Decoder	Any	Learned	Soft	DNN-parameterized; can exceed ML in specific regimes

2.4.2 Hard vs. Soft Decision Decoding

A fundamental tradeoff in FEC implementation is between hard-decision and soft-decision decoding:

Hard-Decision FEC (HD-FEC): The receiver first makes a binary decision (bit is 0 or 1) based on a threshold, then passes the binary sequence to the FEC decoder. The decoder operates on binary data. Advantage: simple, low power, low latency. Disadvantage: discards the analog information about *how close* the received signal was to the threshold, losing approximately 2–3 dB of coding gain compared to soft-decision.

Soft-Decision FEC (SD-FEC): The receiver passes reliability information (log-likelihood ratios, LLRs) directly to the FEC decoder without hard quantization. The decoder uses the probabilistic information in the LLRs to resolve ambiguous bits. Advantage: 2–3+ dB additional NCG; essential for approaching Shannon capacity. Disadvantage: greater hardware complexity, higher power consumption. Used in all modern coherent optical systems (100G and above), 5G NR, and deep-space communications.

2.4.3 Belief Propagation on Tanner Graphs

The Tanner graph is a bipartite graph $G = (V_n, V_m, E)$ representing an LDPC code, where:

- V_n = set of n variable nodes (one per code bit)
- V_m = set of m check nodes (one per parity check equation)
- Edge $(i, j) \in E$ iff column i of H has a 1 in row j

The sum-product algorithm passes messages along edges iteratively. At each check node, the message to variable node i is the estimated log-likelihood that the parity constraint is satisfied given all other variable node messages. At each variable node, the message to check node j is the sum of all incoming check node messages plus the channel LLR for that bit. After I iterations, the decoder makes a hard decision on each bit. The algorithm converges quickly on cycle-free graphs (trees); short cycles in the Tanner graph degrade performance and create the error floor phenomenon.

2.4.4 The Error Floor Problem

All iterative FEC codes (LDPC, turbo) exhibit an "error floor" — a flattening of the BER curve at high E_b/N_0 values where the BER decreases more slowly than the waterfall region. This is caused by:

- **LDPC:** Trapping sets (small absorbing sets in the Tanner graph that cause iterative decoder failures) and near-codewords
- **Turbo codes:** Low-weight codewords corresponding to specific input patterns that the constituent RSC encoders produce

Error floor behavior is critical for optical transport applications, where target BER is 10^{-13} to 10^{-15}, which is well into the error floor regime for many LDPC designs. Staircase codes (a class of spatially coupled LDPC codes) were specifically designed to push the error floor below 10^{-15} and are standardized in ITU-T G.709.2 as the first coherent FEC providing 9.38 dB NCG.

2.4.5 Neural Network Decoders (Emerging)

Deep neural networks are being applied to FEC decoding in three distinct approaches:

1. Deep Learning Decoder (DLD): Replace the belief propagation iterations with a deep unfolded network where each layer corresponds to one BP iteration with learned weights. Demonstrated to approach ML performance with learned weights at shorter block lengths where standard BP underperforms.

2. Model-Free Neural Decoder: Treat the channel model as unknown; train a neural network end-to-end on channel observations and target codewords. Enables

operation on channels for which no analytical model exists.

3. Neural-Assisted BP (NABP): Augment the standard BP algorithm with a neural network that predicts and suppresses trapping set failures, improving the error floor without redesigning the code.

NASA JPL's active research program (Divsalar, Cheng, Dolinar, 2022) applies DNNs specifically to quantum LDPC codes, targeting ML performance with 100%+ throughput improvement and ≥ 2 orders of magnitude reduction in decoding error probability vs. best existing classical decoders.

2.5 Module 04 Reference: Application Domains

2.5.1 Wireless Communications: 2G through 5G NR

The evolution of wireless FEC tracks the progression from simple codes to near-Shannon-limit codes as processing power became available:

Gen.	FEC Scheme	Code Parameters	Target BER / Use Case
2G	Convolutional (CC)	Rate 1/2, K=5	10^{-3} voice; GSM EDGE
3G	Turbo codes (3GPP)	Rate 1/3 (UMTS)	10^{-6} data; HSDPA
4G	Turbo (LTE) + LDPC	Turbo: LTE-specific interleaver; LDPC: 802.11n	10^{-6}–10^{-9}; LTE-Advanced
5G NR	LDPC (data) + Polar (ctrl)	LDPC: BG1 (high rate) / BG2 (low rate); CA-Polar L=8	10^{-5}; eMBB + URLLC

5G NR (3GPP TS 38.212): The 5G New Radio standard uses two distinct FEC schemes:

- **LDPC (transport channels):** Two base graphs (BG1 for large blocks $k > 292$, code rates $R \geq 1/4$; BG2 for small blocks or very low rates). BG1 has maximum block size 8448 bits; BG2 has maximum block size 3840 bits. Quasi-cyclic structure with lift size Z from 2 to 384. Rate matching via shortening and puncturing.
- **Polar (control channels):** CRC-aided polar (CA-Polar) with list size $L = 8$ for PBCH (broadcast), PDCCH (downlink control), and PUCCH (uplink control). Block lengths from 12 to 512 bits.
- **KP4 FEC for PAM4 SerDes:** RS(544,514) with $t = 15$ provides approximately 7 dB coding gain for 200/400GbE interfaces, compensating for the ≈ 9.54 dB detection penalty of PAM4 vs. NRZ signaling.

2.5.2 Optical Fiber Transport: Generations of FEC

Optical transport FEC has evolved through four standardized generations with dramatically increasing NCG:

Generation	Code	Overhead	NCG (dB)	Standard
G-FEC	RS(255,239)	7%	5.6	ITU-T G.709 / G.975
E-FEC	RS(255,223) enhanced	7%	6.2	ITU-T G.975.1
HG-FEC (Staircase)	Staircase HD	20%	9.38	ITU-T G.709.2
SD-FEC (oFEC)	Soft-decision LDPC	20–25%	11.1 (QPSK) / 11.6 (16QAM)	Open ROADM / CableLabs P2P v2.0

The Staircase FEC (HG-FEC), defined in ITU-T G.709.2 and standardized by CableLabs for 100G coherent P2P, provides 9.38 dB NCG at a pre-FEC BER threshold of 4.5×10^{-3}. The openFEC (oFEC) for 200G P2P provides 11.1 dB NCG for QPSK and 11.6 dB for 16QAM at pre-FEC BER of 2×10^{-2}, using 3 soft-decision iterations. Selection between hard-decision and soft-decision FEC involves tradeoffs: SD-FEC provides 1+ dB additional NCG at the cost of higher ASIC complexity, power consumption, and latency.

2.5.3 Deep Space Communications: NASA and CCSDS

Deep space communication represents the most demanding FEC environment: extreme link distances (10^{12} to 10^{13} km), negligible transmit power (a few watts for many probes), and one-way propagation delays of 20 minutes to many hours preventing retransmission. FEC is not optional; it is existential.

Mission / Era	FEC Scheme	Performance	Notes
Voyager 1&2 (1977)	RS(255,223) concatenated with convolutional $R = 1/2$	Near state-of-art for era	Both spacecraft now > 20 billion km distant; FEC still operational
Pioneer 10/11	Simple convolutional $R = 1/2$	Adequate for mission	Early deep-space benchmark
Galileo (1989)	RS(255,223) + CC	Mission saved by FEC	High-gain antenna failed; FEC enabled data recovery at 10 bps
Mars Reconn. Orbiter	Turbo code $R = 1/6$	Near-Shannon at era	CCSDS turbo code; 2–3 dB improvement over previous
Lunar Gateway / LCRD	LDPC + advanced FEC	Approaching Shannon	NASA SCaN program; laser comm. + FEC

NICT/NITech/JAXA 2025 milestone: The first successful demonstration of 5G NR LDPC codes and DVB-S2 LDPC codes on a ground-to-GEO satellite laser communication link (LUCAS satellite) was presented at ICSOS 2025. Using optimized interleaving parameters, both codes successfully corrected burst data errors caused by

atmospheric turbulence fading on a 60 Mbps downlink. This result confirms the viability of terrestrial 5G FEC standards for space optical communication.

2.5.4 Flash Memory and Solid-State Storage

NAND flash memory is the dominant non-volatile storage technology, and its raw BER has been increasing as process nodes shrink and cell densities increase (SLC $\rightarrow$ MLC $\rightarrow$ TLC $\rightarrow$ QLC). FEC is essential to providing the storage-grade reliability ($< 10^{-15}$ unrecoverable bit error rate) that enterprise applications require:

NAND Type	Raw BER	FEC Scheme	Notes
SLC (1 bit/cell)	10^{-6}–10^{-5}	Hamming / BCH (weak)	Sufficient; simple decoder
MLC (2 bits/cell)	10^{-4}–10^{-3}	BCH $t = 8$–24	BCH standard; ECC controller
TLC (3 bits/cell)	10^{-3}–10^{-2}	LDPC (soft decision)	NVMe SSDs; LDPC transition
QLC (4 bits/cell)	10^{-2}–10^{-1}	Strong LDPC + RAID	Higher overhead; shorter P/E cycles

Modern enterprise NVMe SSDs use soft-decision LDPC with dynamic code rate adaptation based on NAND health state. As cells wear out (P/E cycles increase raw BER), the FEC controller reduces the code rate (increases overhead) to maintain the target uncorrectable BER. QLC NAND may require 20–30% ECC overhead in late life.

2.5.5 Video Streaming and Broadcast: Raptor in Production

RaptorQ is the dominant application-layer FEC for broadcast and multicast streaming:

- **ATSC 3.0 (Next Gen TV):** RaptorQ specified in A/331 for both real-time streaming (ROUTE protocol) and file delivery (datacasting). Enables robust mobile TV reception under Rayleigh fading and partial signal blockage.
- **3GPP eMBMS:** Raptor code standardized in 3GPP TS 26.346 for multimedia broadcast multicast service. Qualcomm's analysis (2009) demonstrated that for $k \geq 1000$ source symbols, average reception overhead converges to approximately 0.2% — essentially zero beyond the source symbols.
- **DVB-H and DVB-IPTV:** Raptor codes standardized for handheld TV broadcasting and IP-based delivery.
- **SD-WAN / Enterprise:** Packet-layer FEC (parity over UDP streams) is used for real-time VoIP, video conferencing, and RTP streams. Palo Alto Networks SD-WAN FEC supports branch-to-hub encoding/decoding with per-application error correction profiles. FEC is appropriate only for VPN tunnel links (not DIA links) and only for loss-sensitive applications (VoIP, video), as it introduces bandwidth overhead and CPU cost.

2.6 Module 05 Reference: Emerging Frontiers

2.6.1 Quantum Error Correction (QEC)

Quantum information is inherently fragile: qubits (quantum bits) can suffer bit-flip errors (X errors), phase-flip errors (Z errors), and combined errors ($Y = iXZ$ errors), and are further subject to decoherence and measurement disturbance. Unlike classical bits, quantum information cannot be cloned (no-cloning theorem), making direct redundancy impossible. QEC solves this through the *stabilizer formalism*: encode one logical qubit into multiple physical qubits using entanglement, such that error syndromes (parity checks on the stabilizer generators) can be measured without disturbing the encoded logical information.

Surface Codes (topological codes): The dominant near-term QEC approach. Logical qubits are encoded in a 2D grid of physical qubits. Error correction is achieved by repeated syndrome measurements; the minimum-weight perfect matching (MWPM) decoder decodes the syndrome pattern. The surface code threshold (minimum required physical qubit fidelity for fault-tolerant operation) is approximately 1%, making it compatible with current hardware.

Quantum LDPC (qLDPC) codes: Classical LDPC codes adapted to the quantum setting through the CSS (Calderbank-Shor-Steane) construction, producing codes where the parity check matrix H satisfies the symplectic inner product condition $H_X H_Z^T = 0$. qLDPC codes offer better encoding rates (logical qubits per physical qubit) than surface codes, but have been harder to implement due to long-range qubit connectivity requirements.

2024–2025 Hardware Milestones:

- **Google Willow (December 2024):** First demonstration of QEC clearly bringing errors down in practice as the number of physical qubits increases — the defining threshold behavior proving practical fault tolerance. This was the inflection point where QEC moved from theory to tangible hardware.
- **IBM qLDPC transition (2024):** IBM transitioned production quantum processors to qLDPC codes, achieving better logical qubit fidelity per physical qubit than surface codes at equivalent distances.
- **Oxford Ionics (2025):** Achieved 99.99% fidelity for two-qubit gates using trapped-ion hardware — a threshold result for practical QEC.
- **Research explosion (2025):** 120+ peer-reviewed QEC papers published between January and October 2025, up from 36 in all of 2024 (Riverlane 2025 report). Breaking RSA encryption with quantum computers may require as few as 1 million qubits (down from earlier estimates of 20 million) with improved QEC efficiency.

NASA JPL qLDPC Neural Decoder Research (2022): Active research applying DNNs to decode quantum LDPC codes. Target metrics: approach maximum likelihood (ML) decoder performance with ≥ 100% throughput improvement, ≥ 50% reduction in hardware complexity and power, ≥ 2 orders of magnitude reduction in decoding error probability vs. best existing classical decoders (Divsalar, Cheng, Dolinar et al., JPL Task 2022).

2.6.2 AI-Native and Neural FEC Decoders

The convergence of deep learning and FEC is producing a new class of "learned" decoders that can outperform hand-crafted algorithms in specific regimes:

- **Unfolded BP Networks:** Unfold the I iterations of belief propagation into a I-layer deep network; learn per-layer weights to optimize performance. Particularly effective at short block lengths where standard BP underperforms.
- **Reinforcement Learning for Code Rate Adaptation:** Use RL to dynamically select code rate and overhead in response to channel quality observations, approaching the optimal rate-agile policy.
- **DNN Channel Estimators Interleaved with FEC:** Novel LDPC and polar decoders with integrated DNN channel estimators that iteratively improve channel estimates using soft LLR feedback from the FEC decoder. Demonstrated in the 2023 MDPI Sensors paper on AWGN + ISI channels.
- **End-to-End Learned Coding:** Treat the encoder and decoder as a jointly trained autoencoder over the channel. Has produced near-optimal results for specific short block lengths but scaling to practical block sizes remains an open research problem.

2.6.3 6G Channel Coding Research

The 6G standardization process (targeting commercial deployment ≈ 2030) is evaluating FEC schemes beyond 5G's LDPC + Polar combination. Key directions include:

- **Concatenated RS + Polar codes:** A 2025 *Scientific Reports* (Nature) paper evaluated concatenated RS + next-generation Polar codes for 6G use cases. Polar codes in 6G must address LDPC's limitation at control channels (short block lengths) and Polar's limitation at medium/long block lengths, where list size L must grow large.
- **Semantic coding:** A paradigm shift from Shannon's bit-level reliability toward coding that preserves *meaning* at the application layer, allowing reconstruction of the semantic content of a message even if the precise bit pattern is not recovered. Relevant for AI-generated content, compressed scene representations, and neural radiance field transmission.
- **Ultra-Reliability Low Latency (URLLC) requirements for 6G:** Target 10^{-7} BLER with 100 μs latency at short block lengths. Current polar + LDPC achieve $\approx 10^{-5}$ reliably; 6G targets require 2 orders of magnitude additional reliability improvement.
- **AI-native channel coding:** 3GPP Study Item on AI/ML for NR physical layer (expected standard contributions 2026–2027) includes channel coding as a candidate AI application area.

2.7 Module 06 Reference: GSD / Fox Infrastructure Integration

2.7.1 Pacific Spine Segment FEC Selection

The Fox Infrastructure Group Pacific Spine spans multiple distinct communication segments, each with different noise models and requirements:

Segment Type	Channel Model	Recommended FEC	Rationale
Terrestrial fiber (trunk)	AWGN, very low BER	SD-FEC LDPC (oFEC) 20% OH	Best NCG; latency acceptable
Submarine cable (Pacific)	AWGN + seismic risk	HD Staircase FEC or SD-FEC	Staircase for error floor; SD for long runs
Free-space optical (FSO)	Burst fading (turbulence)	5G NR LDPC + interleaver	NICT 2025 result; burst-error robust
GEO satellite uplink	AWGN + Doppler	DVB-S2 LDPC or turbo $R = 1/3$	Long delay; strong FEC; CCSDS-aligned
LEO constellation link	Fast fading, low latency	Raptor/RaptorQ (erasure)	Topology disruption; fountain robustness
Mesh radio (tribal sites)	Rayleigh fading, burst	5G NR LDPC or RS-based FEC	Short-range; low latency priority
Emergency CSZ mesh	Partial disruption	RaptorQ fountain codes	Topology-agnostic; any-subset recovery
FoxCompute ocean nodes	Storage (NAND)	Strong LDPC (SD, 20–25% OH)	QLC-class raw BER; data integrity critical

2.7.2 DACP Reliability Overlay with FEC Semantics

The GSD Deterministic Agent Communication Protocol (DACP) defines agent-to-agent messages as three-part structured bundles: (1) human intent specification, (2) structured data payload, and (3) executable code block. In multi-hop agent pipelines, these bundles traverse multiple context windows, tool calls, and serialization/deserialization operations, each of which introduces potential for corruption (JSON encoding errors, context truncation, tool call failures).

A DACP reliability overlay informed by FEC principles would include:

- **Structured redundancy:** Bundle checksums (BLAKE3 hash) for integrity verification at each hop; Reed-Solomon or CRC encoding of critical numerical fields (context window IDs, agent addresses, milestone sequence numbers)
- **Erasure-resilient bundle serialization:** Critical agent state encoded in a fountain-code-inspired format where any k-of-n sub-fields contain enough information to reconstruct the full bundle
- **Systematic code structure:** The original plaintext human intent is always present in the bundle (systematic code property), enabling human fallback even when automated decoding fails
- **Interleaving:** For long agent execution chains, interleave bundle fields to distribute context window boundary corruption across multiple fields rather than catastrophically corrupting a single field

- **Soft-decision handoff:** Rather than binary accept/reject of incoming bundles, DACP agents pass confidence-weighted soft assessments to the next agent, enabling turbo-style iterative refinement in multi-agent loops

2.7.3 CSZ Emergency Mesh: Fountain Code Architecture

The Cascadia Subduction Zone (CSZ) emergency mesh must maintain communications when conventional routing fails: fiber cuts, power outages, base station damage, and topology fragmentation. RaptorQ fountain codes are the optimal FEC choice because:

- **Topology-agnostic:** Any receiver that accumulates $\approx k(1+\epsilon)$ packets from any transmission sources (not necessarily sequential or from the same transmitter) can reconstruct the original data. Perfect for disrupted mesh topologies.
- **No coordination required:** The fountain encoding does not require the sender to know which packets the receiver has received. Broadcasting continues indefinitely; receivers decode when they have enough.
- **Near-zero overhead:** Reception overhead $\epsilon \approx 0.002$ for $k \geq 1000$ — only 0.2% more data than the original needed for reconstruction.
- **Linear-time decoding:** Raptor codes decode in $O(k)$ operations, enabling real-time operation on mesh nodes with limited processing power (e.g., Raspberry Pi-class hardware in emergency relay nodes).

2.8 Safety and Sensitivity Considerations

Consideration	Boundary Type	Specification
Source quality	ABSOLUTE	All quantitative claims traced to peer-reviewed papers, standards bodies (IEEE, IETF, ITU-T, 3GPP, CCSDS), or government/agency reports
Numerical attribution	ABSOLUTE	Every NCG value, BER threshold, and overhead percentage attributed to specific standard or paper
No policy advocacy	ABSOLUTE	FEC selection recommendations presented as engineering tradeoffs; no advocacy for specific vendors or proprietary implementations
Security sensitivity	GATE	FEC-cryptography interactions (wiretap channel theory, physical layer security) addressed at survey level only; no operational detail that aids exploitation
Quantum hardware claims	ANNOTATE	QEC hardware results represent state-of-art as of early 2026; quantum computing landscape changes rapidly; appropriate uncertainty language used for forward-looking claims
Tribal/Indigenous site data	ABSOLUTE	Mesh node placement at tribal sovereign sites described architecturally; no specific geographic coordinates published in this document

2.9 Source Bibliography

2.9.1 Foundational Papers

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- Arikan, E. (2009). "Channel Polarization: A Method for Constructing Capacity-Achieving Codes for Symmetric Binary-Input Memoryless Channels." *IEEE Transactions on Information Theory*, 55(7), 3051–3073.

2.9.2 Standards and Technical Specifications

- 3GPP TS 38.212 v15.2.0: "NR; Multiplexing and channel coding" — defines LDPC + Polar for 5G NR.
- ITU-T Recommendation G.709 (2020): "Interfaces for the Optical Transport Network" — defines G-FEC RS(255,239).
- ITU-T Recommendation G.975.1 (2004): "Forward error correction for high bit-rate DWDM submarine systems."
- ITU-T Recommendation G.709.2 (2017): "OTU4 long-reach interface" — defines staircase FEC HG-FEC.
- IETF RFC 6330 (2011): "RaptorQ Forward Error Correction Scheme for Object Delivery." IETF.

- IETF RFC 5053 (2007): "Raptor Forward Error Correction Scheme for Object Delivery." IETF.
- ATSC A/331 (2022): "ATSC 3.0 Signaling, Delivery, Synchronization, and Error Protection." ATSC.
- CCSDS 131.0-B-3 (2019): "TM Synchronization and Channel Coding." CCSDS, Washington D.C.

2.9.3 Peer-Reviewed Research (2019–2025)

- Hussain, G.A. et al. (2025). "Performance analysis of concatenated Reed-Solomon and next generation polar codes for 6G communication systems." *Scientific Reports (Nature)*, 15.
- Frunza, M. et al. (2023). "Performance Analysis of Turbo Codes, LDPC Codes, and Polar Codes over an AWGN Channel in the Presence of Inter Symbol Interference." *Sensors (MDPI)*, 23(4), 1942.
- Divsalar, D., Cheng, M., Dolinar, S. et al. (2022). "Robust Neural Network Decoders for Quantum Error Correction." NASA JPL Research Poster SP22013p.
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STAGE 3 — MISSION PACKAGE

3 Milestone Specification

3.1 Mission Objective

Produce a comprehensive, publication-quality research corpus on Forward Error Correction theory and practice that serves as the foundational reliability engineering reference for the GSD ecosystem, Fox Infrastructure Group Pacific Spine architecture, and DACP protocol reliability layer. The mission will advance from theoretical foundations through cutting-edge quantum/AI frontiers and culminate in actionable FEC architecture specifications for six distinct Fox Infrastructure Group deployment contexts.

3.2 Architecture Overview

```
WAVE EXECUTION OVERVIEW --- FEC RESEARCH MISSION ========================================
WAVE 0: FOUNDATION (Sequential, Haiku/Gary) [ 30min] [Paula] Schema +
templates + bibliography validation [Gary] Source index + shared glossary
definitions Output: Shared context package for all Wave 1 agents
WAVE 1A: PARALLEL RESEARCH TRACK A (3 agents) [ 2hr] [CRAFT-THEORY] Module
01: Theoretical Foundations [CRAFT-CODE] Module 02: Code Families Taxonomy
[CRAFT-DECODE] Module 03: Decoding Algorithms All run simultaneously;
share schema from Wave 0 Output: Three independent research documents
WAVE 1B: PARALLEL RESEARCH TRACK B (2 agents) [ 2hr] [CRAFT-APP] Module
04: Application Domains [CRAFT-FRONTIER] Module 05: Emerging Frontiers
Requires Wave 1A completion for cross-references Output: Two independent
research documents
CAPCOM GATE 1: Human review of Modules 01-05 [ 20min] Scope confirmation +
sensitivity review
WAVE 2: SYNTHESIS (Sequential, Opus) [ 1.5hr] [CRAFT-GSD] Module 06:
GSD/Fox Infrastructure Integration [Agnus] Cross-module integration +
consistency check Consumes all five Wave 1 outputs Output: Integration
document + consistency report
CAPCOM GATE 2: FEC architecture approval [ 15min] Pacific Spine FEC
selection + DACP reliability spec review
WAVE 3: PUBLICATION (Sequential, Denise/Sonnet) [ 1hr] [Denise] Final
LaTeX document assembly [Denise] Three-pass XeLaTeX compilation [Denise]
index.html generation [Gary] Git commit + staging gate Output: PDF + .tex
+ index.html
TOTAL ESTIMATED: 7--8 hours across 6--8 context windows
```

3.3 System Layers

Layer 1: Schema and Glossary Layer (Wave 0): Shared data structures, module output format specs, bibliography schema, and terminology index. Ensures all Wave 1 agents produce interoperable outputs.

Layer 2: Theoretical Foundation Layer (Wave 1A): Three parallel research documents covering the mathematical and algorithmic foundations of FEC. Pure

research; no implementation or architecture specificity.

Layer 3: **Application Intelligence Layer (Wave 1B):** Two parallel documents mapping theoretical foundations to deployed systems and emerging research frontiers.

Layer 4: **Synthesis and Integration Layer (Wave 2):** Module 06 synthesizes all five prior modules into GSD-specific architecture decisions. The cross-module consistency check validates no contradictions between modules.

Layer 5: **Publication Layer (Wave 3):** LaTeX assembly, compilation, index generation, and delivery. Quality gate: document must compile without errors and all success criteria must be verified.

3.4 Deliverables

#	Deliverable	Acceptance Criteria	Component Spec
D-01	Module 01 Research Doc	Shannon theorem + all channel models mathematically defined; ≥ 8 quantitative data points sourced	CRAFT-THEORY output; Opus-level precision
D-02	Module 02 Research Doc	All 7 major code families characterized with (n, k, d_{min}); parameter comparison tables; historical timeline	CRAFT-CODE output; Sonnet-level survey
D-03	Module 03 Research Doc	All 6 decoding algorithms documented; hard vs. soft comparison with dB data; neural decoder section	CRAFT-DECODE output; Sonnet-level survey
D-04	Module 04 Research Doc	8 application domains; standards cross-refs; ≥ 3 case studies; deployment parameters	CRAFT-APP output; Sonnet-level survey
D-05	Module 05 Research Doc	QEC section with 2024–2025 hardware results; AI decoder research; 6G coding research	CRAFT-FRONTIER output; Opus-level judgment
D-06	Module 06 Integration Doc	Pacific Spine FEC table; DACP reliability overlay spec; CSZ emergency mesh spec	CRAFT-GSD + Agnus synthesis; Opus-level
D-07	Compiled PDF	Compiles cleanly; ≥ 40 pages; all 12 success criteria verifiable in text	Denise + Gary (3-pass XeLaTeX)
D-08	LaTeX source (.tex)	Compilable; complete document source; all dependencies documented	Denise; provided alongside PDF
D-09	index.html	Matches color scheme; download links; usage instructions; LaTeX recompilation guide	Denise; domain-appropriate typography

3.5 Component Breakdown

Component	Dependencies	Model	Est. Tokens	Wave
SC-00: Schema	None	Haiku	8,000	0
SC-01: Mod 01 Theory	SC-00	Opus	30,000	1A
SC-02: Mod 02 Codes	SC-00	Sonnet	30,000	1A
SC-03: Mod 03 Decode	SC-00, SC-02	Sonnet	28,000	1A
SC-04: Mod 04 Apps	SC-01, SC-02, SC-03	Sonnet	32,000	1B
SC-05: Mod 05 Frontier	SC-01, SC-02, SC-03	Opus	35,000	1B
SC-06: Mod 06 GSD Integ.	SC-01–SC-05	Opus	40,000	2
SC-07: Consistency	All SC	Agnus/Opu	20,000	2
SC-08: LaTeX Assembly	All SC	Sonnet	18,000	3
SC-09: HTML Index	SC-08	Sonnet	5,000	3
		Total estimated:	**246,000**	

3.6 Model Assignment Rationale

Agent Role	Model	% Budget	Rationale
CRAFT-THEORY	Opus	30%	Foundational mathematics requires deep precision; Shannon theorem derivation; coding gain calculations; judgment about which results are seminal
CRAFT-FRONTIER	Opus	30%	QEC frontier requires calibrated uncertainty; AI decoder research is rapidly evolving; inappropriate to use Sonnet-level pattern matching for frontier claims
CRAFT-GSD (Mod 06)	Opus	30%	Architecture decisions for Fox Infrastructure affect multi-billion dollar infrastructure; conservative judgment required
CRAFT-CODE, CRAFT-DECODE, CRAFT-APP	Sonnet	60%	Survey and synthesis of well-established technical literature; structured table generation; standards cross-reference; appropriate for Sonnet-level pattern matching and synthesis
Paula, Gary (scaffolding)	Haiku	10%	Schema generation, file operations, template filling; low-complexity high-volume tasks

4 Wave Execution Plan

4.1 Wave Summary

Wave	Purpose	Execution	Model Mix	Output
0	Foundation scaffolding	Sequential	Haiku	Shared schema, glossary, source index
1A	3-module parallel research	Parallel (3)	Opus + Sonnet	Modules 01, 02, 03 research docs
1B	2-module parallel research	Parallel (2)	Opus + Sonnet	Modules 04, 05 research docs
2	Synthesis + integration	Sequential	Opus	Module 06 + consistency report
3	Publication + delivery	Sequential	Sonnet	PDF, .tex, index.html

4.2 Wave 0: Foundation

Agent	Task	Output	Spec
Paula (Haiku)	Generate shared schema	schema.json	Defines module output structure, bibliography format, cross-reference IDs
Gary (Haiku)	Build source index	sources.bib	All 25+ sources validated; DOI/IETF RFC/ITU numbers confirmed
Gary (Haiku)	Generate glossary	glossary.md	All FEC terms defined: BER, SNR, NCG, $d_{\min}$, LDPC, QEC, etc.
Paula (Haiku)	LaTeX templates	module_templa	Section headers, table structures, asciibox environments pre-configured

4.3 Wave 1A: Parallel Research Track A

All three agents run simultaneously. Each has access to Wave 0 outputs and the shared source index.

Agent	Model	Module	Deliverable Spec
CRAFT-THEORY	Opus	Mod 01: Theory	Shannon theorem with all math; channel capacity derivation; BER curves; cliff effect analysis; 30k tokens
CRAFT-CODE	Sonnet	Mod 02: Codes	7 code families; parameter tables; historical timeline; code comparison matrix; 30k tokens
CRAFT-DECODE	Sonnet	Mod 03: Decoders	8 decoder algorithms; complexity table; hard/soft comparison; error floor analysis; neural decoder section; 28k tokens

4.4 Wave 1B: Parallel Research Track B

Begins after CAPCOM confirmation that Wave 1A outputs are consistent with the schema. Two agents run simultaneously.

Agent	Model	Module	Deliverable Spec
CRAFT-APP	Sonnet	Mod 04: Apps	8 application domains; deployment tables; 3 case studies; standards cross-refs; 32k tokens
CRAFT-FRONTIER	Opus	Mod 05: Frontier	QEC + hardware milestones; neural decoders; 6G research; calibrated uncertainty on frontier claims; 35k tokens

4.5 Wave 2: Synthesis

Agent	Model	Task	Deliverable Spec
CRAFT-GSD	Opus	Mod 06: GSD Integration	Pacific Spine FEC table (8 segment types); DACP reliability overlay spec; CSZ emergency Raptor spec; Fox Infrastructure rationale; 40k tokens
Agnus	Opus	Consistency check	Cross-module validation; no contradictions in BER values, code parameters, or standards references; consistency report; 20k tokens

4.6 Wave 3: Publication + Verification

Agent	Model	Task	Spec
Denise	Sonnet	LaTeX assembly	Merge all 6 module outputs into master .tex; apply color scheme and templates; 18k tokens
Gary	Haiku	Three-pass compile	`xelatex` $\times 3$; verify no errors; check page count and TOC; fix any compilation issues
Denise	Sonnet	index.html	Color-matched header; download cards for PDF + .tex; usage instructions; 5k tokens
VERIFY	Sonnet	Success criteria check	Verify all 12 SC criteria satisfied; flag any gaps for remediation before delivery
Gary	Haiku	Git staging	Copy to `.planning/staging/inbox/`; commit with DACP-format message

4.7 Cache Optimization Strategy

- **Shared attribute prefix:** All Wave 1 agents receive the identical Wave 0 schema package at the start of their prompts. This prefix is identical and cacheable with 5-minute TTL, reducing token cost by approximately 8,000 tokens across 5 parallel agents.
- **Module output compression:** Each Wave 1 module output is stored as a dense JSON-structured summary alongside the full prose. Wave 2 agents receive the JSON summary first (compressed context) and request full sections as needed.
- **Source index reuse:** The sources.bib file is generated once in Wave 0 and referenced by all subsequent agents, not re-generated per module.
- **LaTeX template prefix:** The LaTeX preamble and command definitions are pre-generated in Wave 0 and provided to Denise as a cached prefix in Wave 3.

4.8 Token Budget

Wave	Agents	Input Tokens	Output Tokens	Total
Wave 0	2 (Haiku)	5,000	8,000	13,000
Wave 1A	3 (Opus+Sonne	45,000	88,000	133,000
Wave 1B	2 (Opus+Sonne	40,000	67,000	107,000
Wave 2	2 (Opus)	60,000	60,000	120,000
Wave 3	3 (Son-net+Haiku)	25,000	23,000	48,000
			Total (est.):	**421,000**

4.9 Dependency Graph

```
FEC MISSION DEPENDENCY GRAPH ==============================
[Wave 0: Schema + Source Index] | +-----+-----+ | | | [01] [02] [03] <--
Wave 1A (parallel) Theory Codes Decoders | | | +--+--+--+--+ | | [04]
[05] <-- Wave 1B (parallel) Apps Frontier | | +--+--+ | [06] <-- Wave 2:
GSD Integration GSD/Fox | [Consistency] <-- Wave 2: Cross-check | [LaTeX
Assembly] <-- Wave 3 | [PDF + HTML] <-- Wave 3 Output
```

5 Test Plan

5.1 Test Categories

Category	Count	Priority	Failure Action	Description
Safety-critical	7	Mandatory	BLOCK	Source quality, numerical attribution, no advocacy, security sensitivity
Core functionality	24	Required	BLOCK	All 12 success criteria, 2 tests per criterion
Integration	8	Required	BLOCK	Cross-module consistency, self-containment, bibliography integrity
Edge cases	6	Best-effort	LOG	Data gaps, frontier uncertainty, temporal currency
Total	**45**			

5.2 Safety-Critical Tests

Test ID	Verifies	Expected Behavior
SC-SRC	Source quality throughout	All numerical claims (NCG values, BER thresholds, overhead percentages, token budgets) traceable to IEEE, IETF, ITU-T, 3GPP, CCSDS, or peer-reviewed journal. Zero entertainment media or blog citations.
SC-NUM	Numerical attribution	Every dB NCG value, BER threshold, and code parameter $(n, k, d_{\min}, t)$ attributed to a specific source document with year.
SC-ADV	No vendor advocacy	Document presents FEC choices as engineering tradeoffs. No recommendation of specific vendor products (Qualcomm, Xilinx, Coherent, etc.) without technical justification.
SC-SEC	Security sensitivity	Physical-layer security (wiretap channel theory) addressed at survey level only. No operational detail that aids interception, jamming, or exploitation.
SC-GEO	Tribal site geography	Pacific Spine mesh node descriptions at tribal sovereign sites contain no specific GPS coordinates, site names, or infrastructure details that could compromise security or sovereignty.
SC-QEC	QEC claims uncertainty	Forward-looking quantum claims (qubit counts, timeline projections, commercial viability) include explicit uncertainty language. Hardware results from 2024–2025 cited to named organizations.
SC-DACP	DACP spec alignment	DACP reliability overlay specification consistent with GSD DACP milestone (v1.49+) three-part bundle structure. No changes to DACP core semantics proposed without explicit CAPCOM approval.

5.3 Core Functionality Tests (Selected)

Test ID	Success Criterion	Verification Method
CF-01	SC-1: Shannon theorem	Mathematical statement of channel capacity with all variables defined appears in Module 01; tradeoff quantitatively described
CF-02	SC-1: BER characterization	At least 3 BER data points (Shannon limit, turbo codes, LDPC) with sources cited
CF-03	SC-2: Code families complete	All 7 code families (Hamming, BCH, RS, Conv., Turbo, LDPC, Polar) appear in parameter table with $(n, k, d_{\text{min}}, t, R)$
CF-04	SC-2: Historical timeline	Timeline in Module 02 covers 1948–2025 with no major omission
CF-05	SC-3: RaptorQ spec	RFC 6330 cited; 56,403 source symbol parameter documented; ATSC 3.0 deployment noted
CF-06	SC-3: Rateless property	Fountain/rateless property explained with reception overhead formula
CF-07	SC-4: Decoder taxonomy	Table maps each code family to canonical decoder with complexity class
CF-08	SC-4: Hard/soft comparison	2–3 dB soft-decision gain documented with source
CF-09	SC-5: 5G NR coding	BG1/BG2 LDPC documented; CA-Polar $L = 8$ documented; KP4 FEC documented
CF-10	SC-6: Deep space timeline	Voyager RS, Galileo rescue, Mars turbo, LCRD all documented
CF-11	SC-7: Optical FEC generations	G-FEC, E-FEC, HG-FEC, SD-FEC documented with NCG values in dB
CF-12	SC-8: QEC hardware results	Google Willow Dec 2024, IBM qLDPC, Oxford Ionics 2025 all cited
CF-13	SC-9: Neural decoder research	JPL DNN decoder, at least 2 other neural FEC results documented
CF-14	SC-10: Pacific Spine FEC table	6+ segment types mapped to FEC recommendations in Module 06
CF-15	SC-11: DACP reliability spec	Conceptual DACP reliability overlay architecture produced
CF-16	SC-12: CSZ emergency spec	Fountain code recommendation for CSZ mesh with technical justification

5.4 Verification Matrix

Success Criterion	Test ID(s)	Status
SC-1: Shannon theorem + BER characterization	CF-01, CF-02	Pending
SC-2: 7 code families with parameters	CF-03, CF-04	Pending
SC-3: Fountain/Raptor/RaptorQ documentation	CF-05, CF-06	Pending
SC-4: Decoder taxonomy + hard/soft comparison	CF-07, CF-08	Pending
SC-5: 5G NR channel coding standard	CF-09	Pending
SC-6: Deep space FEC timeline	CF-10	Pending
SC-7: Optical transport FEC generations	CF-11	Pending
SC-8: Quantum error correction documentation	CF-12	Pending
SC-9: AI/neural decoder research	CF-13	Pending
SC-10: Pacific Spine FEC selection guide	CF-14	Pending
SC-11: DACP reliability layer specification	CF-15	Pending
SC-12: CSZ emergency mesh specification	CF-16	Pending

6 Mission Crew Manifest

Role	Agent	Model	Responsibility
FLIGHT	Agnus	Opus	Overall mission direction; go/no-go gates; cross-module integration
PLAN	Planner	Sonnet	Wave decomposition; parallel track optimization; token budget management
SCOUT	Research Agent	Sonnet	Gap-fill web research; source validation; bibliography maintenance
EXEC-A	Denise	Sonnet	Document assembly, LaTeX compilation, HTML generation
EXEC-B	Gary (68000)	Haiku	File operations, compilation runs, git operations, schema generation
CRAFT-THEORY	Module 01 Specialist	Opus	Shannon theorem, channel models, information theory mathematics
CRAFT-CODE	Module 02 Specialist	Sonnet	Code families taxonomy, parameters, historical timeline
CRAFT-DECODE	Module 03 Specialist	Sonnet	Decoding algorithms, complexity analysis, error floor, neural decoders
CRAFT-APP	Module 04 Specialist	Sonnet	Application domains, standards, case studies, deployment parameters
CRAFT-FRONTIER	Module 05 Specialist	Opus	QEC, AI/ML decoders, 6G research, frontier uncertainty management
CRAFT-GSD	Module 06 Specialist	Opus	GSD/Fox Infrastructure integration, DACP spec, CSZ emergency spec
VERIFY	Verification Agent	Sonnet	Source validation; accuracy checking; success criteria verification
INTEG	Integration Agent	Sonnet	Cross-module relationship validation; consistency checks
CAPCOM	HITL Interface	Human	Human approval at wave boundaries; scope confirmation; sensitivity review
BUDGET	Resource Manager	Haiku	Token budget tracking; session planning; cache policy monitoring
SURGEON	Health Monitor	Haiku	Research quality degradation detection; agent telemetry
LOG	Chronicle Agent	Haiku	Audit trail; DACP-format commit messages; milestone summaries
TOPO	Topology Manager	Sonnet	Agent team structure; mid-mission restructuring if scope changes
ARCH	Architecture Specialist	Opus	Cross-cutting structural decisions; DACP spec review
SECURE	Security Warden	Sonnet	Safety boundary enforcement; SC-SEC and SC-GEO compliance

7 Execution Summary

Metric	Value
Activation Profile	Fleet (20 roles)
Research Modules	6 (Foundations, Codes, Decoders, Applications, Frontiers, GSD Integration)
Research Passes Completed	5 (Foundations, Code Families, Applications/Space, Distributed/Mesh, Fountain/Streaming)
Wave Count	4 (0: Foundation, 1A: 3-parallel, 1B: 2-parallel, 2: Synthesis, 3: Publication)
Parallel Tracks	Wave 1A: 3 agents; Wave 1B: 2 agents
Estimated Token Budget	421,000 tokens total
Model Split	Opus 30% / Sonnet 60% / Haiku 10%
Estimated Context Windows	18–24 across 6–8 sessions
Safety-Critical Tests	7 (all MANDATORY PASS / BLOCK)
Total Tests	45 (7 safety + 24 core + 8 integration + 6 edge)
Success Criteria	12 (all required for delivery acceptance)
GSD Milestone Target	v1.50 research companion (Space Between Ch. 7–8: Information Theory)
Key Deliverables	PDF + .tex + index.html + consistency report
Fox Infrastructure Outputs	Pacific Spine FEC table; DACP reliability overlay; CSZ emergency spec
Primary Domain Authorities	IEEE, IETF, ITU-T, 3GPP, CCSDS, NASA/JPL, Nature/Scientific Reports

8 Closing

On Reliability and Structure

"The fundamental problem of communication is that of reproducing at one point either exactly or approximately a message selected at another point."

— Claude Shannon, *A Mathematical Theory of Communication*, 1948

Shannon's definition of "the fundamental problem" is deceptively simple. It took the entire field of coding theory 75 years to produce codes that solve it optimally. Hamming gave us the geometry. Reed-Solomon gave us polynomial resilience. Gallager gave us sparse graphs that nobody noticed for 36 years. Berrou gave us the turbo revolution. Gallager's codes came back as LDPC. Arikan polarized the channel and proved capacity was achievable constructively. Shokrollahi made codes rateless.

Each breakthrough was not more redundancy — it was better architecture. Not louder, but smarter. Not more bits, but better structure. The spaces between valid codewords carry more information about what was sent than the codewords themselves, because the geometry of those gaps is exactly what the channel has to distort to corrupt a message beyond recovery.

For the GSD ecosystem, this is the deepest implementation of the Amiga Principle available: the entire field of reliable communication is a 75-year proof that architectural elegance, not raw power, is what closes the gap between theory and reality. The Pacific Spine will be built on this proof.

— GSD Research Mission Package, FEC Deep Research, March 2026